1. Functions of Several Variables

1.1. Introduction

We will mostly be working with functions of the form $f(x, y)$ or $g(u, v)$, etc., which take two real values as inputs and produce a unique real value as output.

If a function f is defined for $(x, y) \in D \subseteq \mathbb{R}^2$ then we write

$$f : D \to \mathbb{R}.$$

- D is called the domain of f.
- In the above example $\mathbb{R}$ is the codomain of f; it is the set to which all outputs of f must belong.
- The range (or image) of f is the set of values in the codomain which are actual outputs of f, i.e.

$$\text{range}(f) = \{z \in \text{codomain}(f) : \exists (x, y) \in \text{domain}(f) \text{ s.t. } z = f(x, y)\}$$

Example 1.1

Consider $f(x, y) = \sqrt{x^2 + y}$. What is the largest possible domain $D \subset \mathbb{R}^2$ for f? What is a sensible codomain for f? What is its range?

1.2. Limits

We say $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ has the limit L as (x, y) approaches (x_0, y_0):

$$\lim_{(x,y)\to(x_0,y_0)} f(x, y) = L$$

if when (x, y) approaches (x_0, y_0) along ANY path in the domain D, $f(x, y)$ gets close to L.

Note

1. The limit can exist if f is undefined at (x_0, y_0).
2. L must be finite.

Laws of Limits

If c, L, M are finite real constants, and we have

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L \quad \text{and} \quad \lim_{(x,y)\to(x_0,y_0)} g(x,y) = M,$$

then

1. $\displaystyle\lim_{(x,y)\to(x_0,y_0)} [f(x,y) + g(x,y)] = L + M$
2. $\displaystyle\lim_{(x,y)\to(x_0,y_0)} [cf(x,y)] = cL$
3. $\displaystyle\lim_{(x,y)\to(x_0,y_0)} [f(x,y)g(x,y)] = LM$
4. $\displaystyle\lim_{(x,y)\to(x_0,y_0)} \left[\frac{f(x,y)}{g(x,y)}\right] = \frac{L}{M}$ if $M \neq 0$.

Example 2.1

Evaluate $\lim_{(x,y)\to(0,1)} \dfrac{x+3}{5xy-y^3}$.

Note

If we cannot apply tricks to evaluate the limit, then we try examining $\lim\limits_{(x,y)\to(x_0,y_0)} f(x,y)$ along different paths approaching (x_0, y_0).

Example 2.2

Evaluate $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2}{x^2+y^2}$.

Note that it is of the form $\frac{0}{0}$ at $(0,0)$.

Example 2.3

Evaluate $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy}{x^2+y^2}$.

Note that it is of the form $\frac{0}{0}$ at $(0,0)$.

Example 2.4

Evaluate $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2}{\sqrt{x^2+y^2}}$.

Note that it is of the form $\frac{0}{0}$ at $(0,0)$.

Sandwich Theorem

Theorem 2.5

Suppose for (x, y) *near* (x_0, y_0) *that* f, g, *and* h *are functions such that*

$$g(x, y) \leq f(x, y) \leq h(x, y).$$

Then if $\lim_{(x,y)\to(x_0,y_0)} g(x, y) = L$ *and* $\lim_{(x,y)\to(x_0,y_0)} h(x, y) = L$, *then*

$$\lim_{(x,y)\to(x_0,y_0)} f(x, y) = L.$$

Note

In practice, g and h are chosen to be continuous functions (see definition later) so that their limits are easy to compute.

Note

Here we are not being very precise about what "near" means. What do you think it should mean?

Returning to Example 2.4:

1.3. Continuity

We say that f is continuous at $(x, y) = (x_0, y_0)$ if

$$\lim_{(x,y)\to(x_0,y_0)} f(x, y) = f(x_0, y_0).$$

Note

Note that f must be defined at (x_0, y_0), so (x_0, y_0) must be in the domain of f.

Example 3.1

Is f, given by $f(x, y) = \dfrac{x^2}{x^2 + y^2}$, continuous at (0,0)?

Theorem 3.2

The following function types are continuous at every point in their domains: polynomials, trigonometric functions, nth root functions, exponentials, logarithms, hyperbolic functions.

Theorem 3.3

If f and g are continuous at (x_0, y_0) and $c \in \mathbb{R}$, then the following functions are continuous at (x_0, y_0):

1. $f + g$
2. cf
3. fg
4. $\dfrac{f}{g}$ *if* $g(x_0, y_0) \neq 0$
5. $h \circ f$ *where* h *is continuous at* $z = f(x_0, y_0)$. *Here* $(h \circ f)(x, y) = h(f(x, y))$ *and* $h : D \to \mathbb{R}$ *where* $D \subseteq \mathbb{R}$.

Example 3.4

Where is f, given by

$$f(x, y) = \log(1 - xy),$$

continuous?

Example 3.5

Where is f, given by

$$f(x,y) = \begin{cases} \dfrac{x^2}{\sqrt{x^2+y^2}}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0). \end{cases}$$

continuous?

1.4. Differentiability

We say that f is differentiable at (x_0, y_0) if

1. the partial derivatives $\dfrac{\partial f}{\partial x}$, $\dfrac{\partial f}{\partial y}$ exist at (x_0, y_0); and
2. a tangent plane

$$z = f(x_0, y_0) + \left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} (x - x_0) + \left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} (y - y_0)$$

is a good approximation to $f(x, y)$ at (x_0, y_0), that is,

$$\lim_{(x,y)\to(x_0,y_0)} \frac{f(x, y) - f(x_0, y_0) - \boldsymbol{\nabla} f(x_0, y_0) \cdot (x - x_0, y - y_0)}{\sqrt{(x - x_0)^2 + (y - y_0)^2}} = 0$$

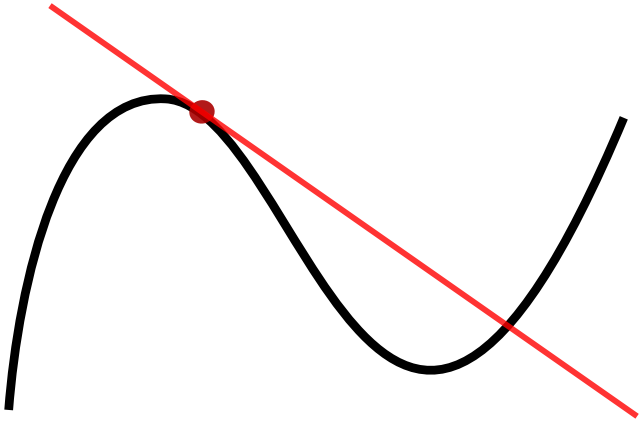
where $\boldsymbol{\nabla} f$ is the gradient of f

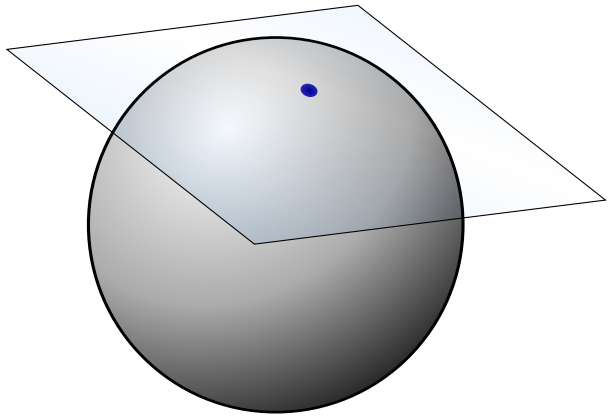
$$\boldsymbol{\nabla} f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$$

and

$$\boldsymbol{\nabla} f(x_0, y_0) = \left.\boldsymbol{\nabla} f(x, y)\right|_{(x,y)=(x_0,y_0)}.$$

The tangent plane is the generalisation of a tangent line for a function of one variable:





Then the equivalent of Condition 2. would say that the tangent line

$$y = f(x_0) + f'(x_0)(x - x_0)$$

should be a good approximation to $f(x)$ at $x = x_0$:

$$\lim_{x \to x_0} \frac{f(x) - f(x_0) - f'(x_0)(x - x_0)}{x - x_0} = 0.$$

Theorem 4.1

If $\dfrac{\partial f}{\partial x}, \dfrac{\partial f}{\partial y}$ *exist and are continuous at* (x_0, y_0) *then* f *is differentiable at* (x_0, y_0).

We will sometimes use the shorthand

$$f_x = \frac{\partial f}{\partial x} \quad \text{and} \quad f_y = \frac{\partial f}{\partial y}$$

and similarly

$$f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$

etc.

Example 4.2

Where is f, given by

$$f(x, y) = x^2 + y^2,$$

differentiable?

Example 4.3

Where is f, given by

$$f(x,y) = \begin{cases} \dfrac{x^2}{\sqrt{x^2+y^2}} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0), \end{cases}$$

differentiable?

Gradient vector and level set

We find another geometric meaning to the vector $\nabla f(x_0, y_0)$, the gradient vector of f at (x_0, y_0).

For any number $c \in \mathbb{R}$, the set $\{(x, y) \mid f(x, y) = c\}$ is called a level set of f with value c.

Example 4.4

Consider Example 4.2, and draw a few level sets of the function $f(x, y) = x^2 + y^2$.

Which level set does the point $(1, 0)$ lie on? What can you say about the tangent line to this level set? What can you say about the gradient and the level set?

Order C^n

A function is C^n if all of its n-th order partial derivatives exist and are continuous.

If a function is C^n, it is automatically $C^0, C^1,\ C^2,\ C^3, \ldots,\ C^{n-1}$.

Example 4.5

f is C^1 if $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ exist and are continuous.
$\Rightarrow$ All C^1 functions are differentiable.

Note

If a function is C^n then the order of differentiation in the partial derivatives of order n is NOT important.

Example 4.6

If f is C^2 then

$$f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = f_{xy}$$

Example 4.7

Where is

$$f(x, y) = \left(x^2 + y^2\right)^{\frac{3}{4}},$$

C^1?

1.5. Chain rule for functions of several variables

Example 5.1

Let $h = h(u, v, w) : \mathbb{R}^3 \to \mathbb{R}$ where $u = u(x, y, z)$, $v = v(x, y, z)$, $w = w(x, y, z)$.
Find $\dfrac{\partial h}{\partial x}, \dfrac{\partial h}{\partial y}, \dfrac{\partial h}{\partial z}$.

Note

For change of variables in multiple integrals we need the Jacobian, the determinant of the Jacobi Matrix.

$$\text{Jacobian} = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \det \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{bmatrix}$$

Inverse Function Theorem

Suppose that $\dfrac{\partial(u,v,w)}{\partial(x,y,z)} \neq 0$.

Then the linear equations relating the changes in coordinates can be inverted and

$$\begin{bmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} & \dfrac{\partial u}{\partial z} \\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} & \dfrac{\partial v}{\partial z} \\ \dfrac{\partial w}{\partial x} & \dfrac{\partial w}{\partial y} & \dfrac{\partial w}{\partial z} \end{bmatrix} \begin{bmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} & \dfrac{\partial x}{\partial w} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} & \dfrac{\partial y}{\partial w} \\ \dfrac{\partial z}{\partial u} & \dfrac{\partial z}{\partial v} & \dfrac{\partial z}{\partial w} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Taking determinants, we get

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} \, \frac{\partial(x,y,z)}{\partial(u,v,w)} = 1.$$

Derivative Matrix

If $f : \mathbb{R}^n \to \mathbb{R}^m$ is a differentiable function, then the derivative is an $m \times n$ matrix given by:

$$\mathbf{D}f = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

where

$$f(x_1, \ldots, x_n) = (f_1(x_1, \ldots, x_n), f_2(x_1, \ldots, x_n), \ldots, f_m(x_1, \ldots, x_n)) .$$

Example 5.2

Find the derivative matrix for the function f defined by

$$f: \mathbb{R}^2 \to \mathbb{R}^3, \ f(u, v) = (u + v, u, v^2).$$

Example 5.3

Find the derivative matrix for the function g defined by

$$g : \mathbb{R}^2 \to \mathbb{R}^2,\ g(x, y) = (x^2 + 1, y^2).$$

Matrix Version of Chain Rule

If $f : \mathbb{R}^m \to \mathbb{R}^p$ and $g : \mathbb{R}^n \to \mathbb{R}^m$ are differentiable functions and the composition $f \circ g$ is defined then,

$$\mathbf{D}(f \circ g) \;=\; \mathbf{D}f\,\mathbf{D}g.$$

We make the variables explicit:

$$\mathbf{D}(f \circ g)(x_1, \ldots, x_n) \;=\; \mathbf{D}f(g(x_1, \ldots, x_n)))\,\mathbf{D}g(x_1, \ldots, x_n).$$

Compare with the chain rule in the 1-variable case.

Note

$$(f \circ g)(x_1, x_2, \ldots, x_n) \;=\; f(g(x_1, x_2, \ldots, x_n)).$$

Example 5.4

Let $f(u, v) = (u + v, u, v^2)$ and $g(x, y) = (x^2 + 1, y^2)$. Compute $\mathbf{D}(f \circ g)$ at $(x, y) = (1, 1)$.

Example 5.5

Let $f(u, v, w) = u^2 + v^2 - w$ and $g(x, y, z) = (x^2 y, y^2, e^{-xz})$.
Compute $\mathbf{D}(f \circ g)$ at $(x, y, z) = (0, 1, 2)$.

Example 5.6

Let $f(x, y) = (x^2, 2x + y, y^3)$ and $g(u, v, w) = (u^2 + 2w, u - v^2)$. Find the derivative of $f(g[f(x, y)])$ at $(1, 0)$.

1.6. Taylor polynomials

Recall the Taylor polynomial for a function $g(x)$ near $x = a$:

$$T(x) = g(a) + g'(a)(x-a) + \frac{g''(a)}{2!}(x-a)^2 + \frac{g'''(a)}{3!}(x-a)^3 + \dots$$

If a function f in two variables is of order C^n, we can approximate $f(x, y)$ by a polynomial of order n around (a, b).

- Approximate $f(x, y)$ by a linear function $p_1(x, y)$ near (a, b) (i.e. $n = 1$)

$$\begin{aligned} p_1(x,y) = &\ f(a,b) + \left.\frac{\partial f}{\partial x}\right|_{(a,b)}(x-a) + \left.\frac{\partial f}{\partial y}\right|_{(a,b)}(y-b) \\ = &\ f(a,b) + (x-a, y-b) \cdot \left.\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)\right|_{(a,b)} \\ = &\ f(a,b) + (x-a, y-b) \cdot \nabla f(a,b). \end{aligned}$$

• The nth order Taylor polynomial for a function $f(x, y)$ of two variables, at (a, b) is

$$p_n(x, y) = \sum_{0 \le k \le n} \sum_{0 \le i \le k} \frac{(x-a)^i (y-b)^{k-i}}{i!(k-i)!} \frac{\partial^k f}{\partial x^i \partial y^{k-i}}(a, b).$$

Here $0! = 1$, $m! = 1 \cdot 2 \cdot 3 \cdots m$.

Expanding this expression when $n = 3$ gives the third order Taylor polynomial for $f(x, y)$ at (a, b):

$$
\begin{aligned}
p_3(x, y) &= f(a, b) + (x - a)f_x(a, b) + (y - b)f_y(a, b) \\
&+ \frac{1}{2}\left[(x - a)^2 f_{xx}(a, b) + 2(x - a)(y - b)f_{xy}(a, b) + (y - b)^2 f_{yy}(a, b)\right] \\
&+ \frac{1}{6}\left[(x - a)^3 f_{xxx}(a, b) + 3(x - a)^2(y - b)f_{xxy}(a, b)\right. \\
&+ \left. 3(x - a)(y - b)^2 f_{xyy}(a, b) + (y - b)^3 f_{yyy}(a, b)\right]
\end{aligned}
$$

Taylor's Theorem for Remainder

The truncation error or remainder $R_n(x, y)$ when approximating $f(x, y)$ around (a, b) by the nth order Taylor polynomial $p_n(x, y)$ at (a, b) is

$$R_n(x, y) = f(x, y) - p_n(x, y).$$

Suppose that f is C^{n+1}. Taylor's formula for the remainder is

$$\begin{aligned} &R_n(x, y) = f(x, y) - p_n(x, y) \\ &= \sum_{0 \le i \le n+1} \frac{(x-a)^i (y-b)^{n+1-i}}{i!(n+1-i)!} \frac{\partial^{n+1} f}{\partial x^i \partial y^{n+1-i}} \left(a + \xi(x-a), b + \xi(y-b)\right) \end{aligned}$$

for some $0 < \xi < 1$.

For example, the error in the approximation using a 2nd order Taylor polynomial is

$$
\begin{aligned}
&R_2(x,y) = f(x,y) - p_2(x,y) \\
&= \frac{1}{6}\left[\Delta_x^3 f_{xxx}(a+\xi\Delta_x, b+\xi\Delta_y) + 3\Delta_x^2\Delta_y f_{xxy}(a+\xi\Delta_x, b+\xi\Delta_y)\right. \\
&\left. + 3\Delta_x\Delta_y^2 f_{xyy}(a+\xi\Delta_x, b+\xi\Delta_y) + \Delta_y^3 f_{yyy}(a+\xi\Delta_x, b+\xi\Delta_y)\right]
\end{aligned}
$$

for some $0 < \xi < 1$, where $\Delta_x = x - a$ and $\Delta_y = y - b$.

Example 6.1

Find the 2nd order Taylor polynomial for $f(x, y) = e^{xy}$ near $(1, 0)$. Hence, approximate $e^{0.11}$.

Example 6.2

Find the linear Taylor polynomial $p_1(x, y)$ and quadratic Taylor polynomial $p_2(x, y)$ for $f(x, y) = \frac{1}{x^2 y}$ near $(1, 1)$. Compare $p_1(0.9, 1, 1)$, $p_2(0.9, 1, 1)$, and $f(0.9, 1.1)$.

Example 6.3

Find the 2nd order Taylor polynomial for $f(x, y) = \sin(x + 2y)$ near $(0, 0)$. Find an upper bound for the error if $|x| < 0.1$ and $|y| < 0.1$.

Example 6.4

Evaluate $\lim_{(x,y)\to(0,0)} \dfrac{\sin(x+2y)}{x+2y}$.

1.7. Extrema

A point (a, b) is called a critical point of $f(x, y)$ if

- either $\nabla f(a, b) = \mathbf{0}$
- <u>OR</u> ∇f does not exist at (a, b).

The first derivative test says if $f(x, y)$ has a local maximum or local minimum at (a, b) and ∇f exists at (a, b), then $\nabla f(a, b) = \mathbf{0}$.

Near a critical point (a, b) such that $\nabla f(a, b) = \mathbf{0}$, the 2nd order Taylor polynomial reduces to

$$\begin{aligned} p_2(x, y) = f(a, b) + \ & \frac{1}{2}[(x-a)^2 f_{xx}(a, b) \\ & + 2(x-a)(y-b) f_{xy}(a, b) + (y-b)^2 f_{yy}(a, b)]. \end{aligned}$$

$$\Rightarrow \ p_2(x, y) \ = \ f(a, b) + \frac{1}{2} \begin{bmatrix} x-a & y-b \end{bmatrix} \begin{bmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{xy}(a, b) & f_{yy}(a, b) \end{bmatrix} \begin{bmatrix} x-a \\ y-b \end{bmatrix}$$

Let

$$H(x, y) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{bmatrix}$$

called the Hessian matrix. Then

$$\det(H) = f_{xx} f_{yy} - (f_{xy})^2 .$$

The Hessian determines whether the critical points are local maxima, local minima or saddle points.

The second derivative test says if (a, b) is a critical point of $f(x, y)$ such that $\nabla f(a, b) = \mathbf{0}$, then

1. $\det H(a, b) > 0$ and $f_{xx}(a, b) < 0 \Rightarrow$ local maximum at (a, b)
2. $\det H(a, b) > 0$ and $f_{xx}(a, b) > 0 \Rightarrow$ local minimum at (a, b)
3. $\det H(a, b) < 0 \Rightarrow$ saddle point at (a, b)
4. $\det H(a, b) = 0 \Rightarrow$ test is inconclusive.

Note

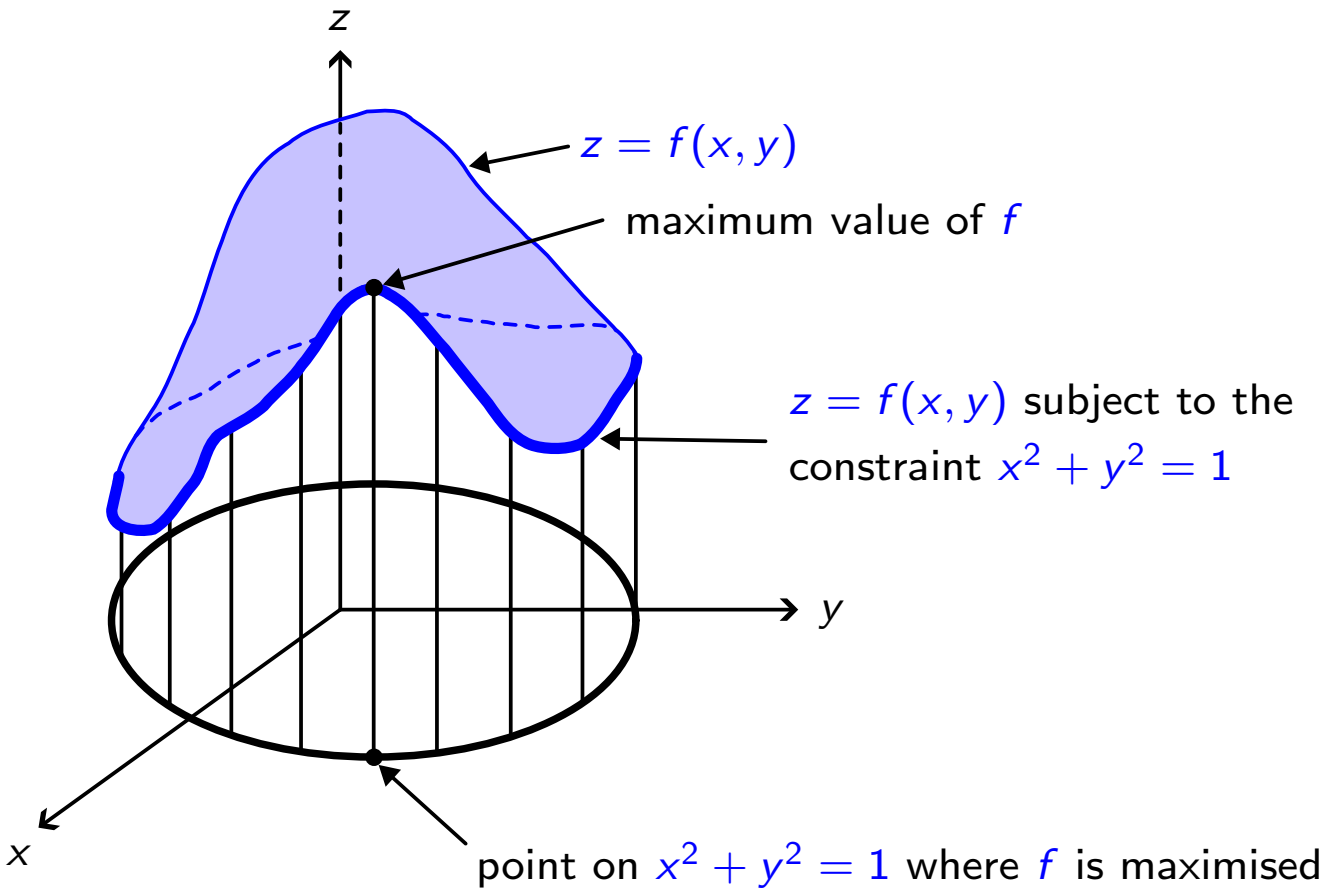
Suppose that (a, b) is a critical point of $f(x, y)$ such that $\nabla f(a, b) = \mathbf{0}$. Using the language of Linear Algebra,

1. if $H(a, b)$ is positive definite, then f has a local minimum at (a, b);
2. if $H(a, b)$ is negative definite, then f has a local maximum at (a, b);
3. if $H(a, b)$ is indefinite and non-singular, then f has a saddle point at (a, b);
4. if $H(a, b)$ is singular, then the test is inconclusive.

Constrained extrema

Suppose we wish to find the maximum and minimum values of a function of several variables subject to one or more constraints.

For example, if we wish to maximise a function of two variables $f(x, y)$ subject to the constraint $x^2 + y^2 = 1$:



Guidelines for Extrema

A set D is <u>bounded</u> if there is a finite real number M such that $|x| < M$ for all $x \in D$.

A set is <u>closed</u> if it contains all of its boundary points.

Theorem 7.1 (Extreme value theorem)

If $f(x, y)$ is continuous on a closed bounded set in $\mathbb{R}^2$, then f attains an absolute maximum value at some point (x_1, y_1) and an absolute minimum value at some point (x_2, y_2), where (x_1, y_1) and (x_2, y_2) are either a boundary point or a critical point.

Note

If the set is unbounded or it is not closed, then maxima and minima need not exist.

Example 7.2

Closed, bounded set.

$$x^2 + y^2 + z^2 = 1$$

Example 7.3

Bounded set, not closed.

$$x^2 + y^2 + z^2 < 1$$

Example 7.4

Closed, unbounded constraint.

$$x + y = 1$$

To find absolute maximum/minimum of $f(x, y)$ on a closed and bounded set D

1. find values of f on critical points
2. find values of f on boundary points
3. compare.

Example 7.5

Find absolute maximum/minimum of $f(x, y) = xy - x - y + 3$ on the triangle with vertices $(0, 0), (2, 0), (0, 4)$.

1.8. Lagrange Multipliers

Example 8.1

Find maximal and minimal values of $f(x, y) = x^2 + y^2$ subject to the condition $\frac{x^2}{4} + y^2 = 1$.

Theorem 8.2

Let f and g be C^1 functions. If $\mathbf{a}$ is an extremum of f subject to the constraint $g(\mathbf{x}) = 0$, then either there exists $\lambda \in \mathbb{R}$ (called a Lagrange multiplier) such that

$$\nabla f(\mathbf{a}) = \lambda \nabla g(\mathbf{a})$$

or

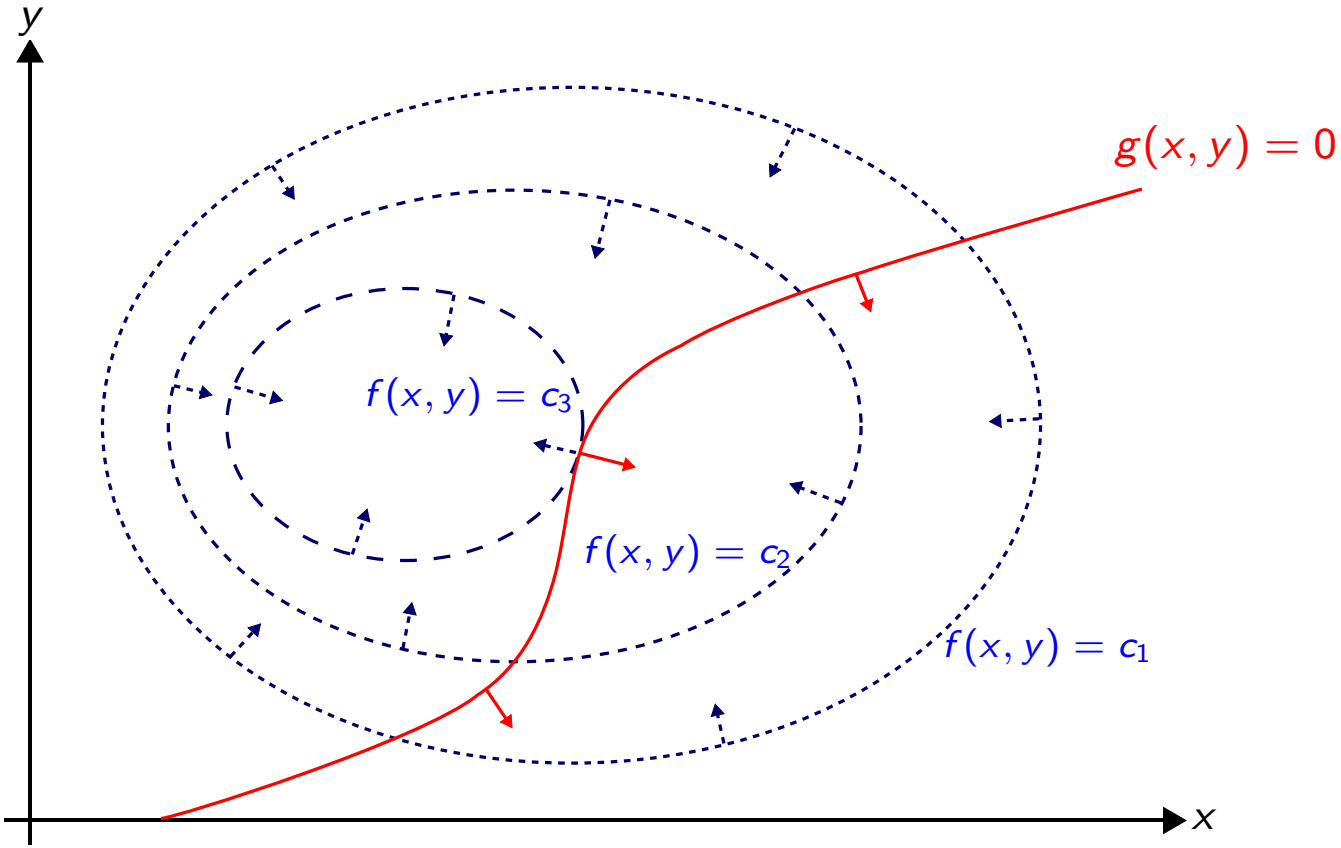
$$\nabla g(\mathbf{a}) = \mathbf{0}.$$

Geometrically

- $\nabla f(\mathbf{a})$ is normal to a level set of f.
- $\nabla g(\mathbf{a})$ is normal to $g(\mathbf{x}) = 0$.
- At the critical points $\nabla f(\mathbf{a})$ and $\nabla g(\mathbf{a})$ are parallel.

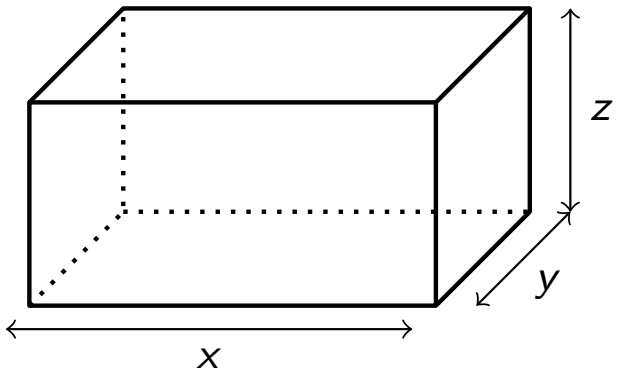
So the constraint $g(\mathbf{x}) = 0$ will be tangent to a level set of f at the critical points.

Visualising:



Example 8.3

An open rectangular box is to be made with fixed volume of $4m^3$. What dimensions should the box have to minimise the amount of material used to make it?



Theorem 8.4

Let f, g_1 *and* g_2 *be* C^1 *functions.*

Suppose that $\mathbf{a}$ *is an extremum of* f *subject to the constraints*

$$g_1(\mathbf{x}) = 0 \text{ and } g_2(\mathbf{x}) = 0,$$

with

$$\nabla g_1(\mathbf{a}) \text{ and } \nabla g_2(\mathbf{a})$$

linearly independent.

Then there exist Lagrange multipliers $\lambda_1, \lambda_2 \in \mathbb{R}$ *such that*

$$\nabla f(\mathbf{a}) = \lambda_1 \nabla g_1(\mathbf{a}) + \lambda_2 \nabla g_2(\mathbf{a}).$$

Example 8.5

Find the extrema of f given by

$$f(x, y, z) = x + y + z$$

subject to the conditions

$$x^2 + y^2 = 2 \text{ and } x + z = 1.$$

Example 8.6

Find the absolute maximum and minimum of f, given by

$$f(x, y) = \frac{1}{2}(x^2 + y^2)$$

in the region $\frac{1}{2}x^2 + y^2 \leq 1$.

Geometrical Interpretation